

Exact Clocks and Time Reversal in Dyadic Pascal Skew Towers

Route-A structural certificate C166

Abstract

For $q = 2^r$ and every $d \geq 2$, consider the finite affine tower $T(x_1, \dots, x_d) = (x_1 + x_2, \dots, x_{d-1} + x_d, x_d + 1)$. With $a = \lfloor \log_2 d \rfloor$ and $M = 2^{r+a}$, we prove that $\text{Fix}(T^n)$ is the entire q^d -state space exactly when $M \mid n$, and is empty otherwise. Thus every state has least period M , there are q^d/M primitive cycles, and $\zeta_T(z) = (1 - z^M)^{-q^d/M}$. The finite Koopman determinant is its reciprocal. In truncated-polynomial coordinates, T is multiplication by $1 + t$ and substitution $t \mapsto -t/(1 + t)$ is an involutive reversor, yielding an exact antiunitary time reversal. The proof uses Pascal coefficients and a sharp binary-valuation witness; finite sweeps are only regression sentinels. No complexity, target determinant, arithmetic, or Hilbert–Pólya claim is made.

Keywords: affine dynamics; Pascal tower; dyadic valuation; exact period; Artin–Mazur zeta; Koopman unitary; time reversal.

中文摘要：本文研究定义在有限模空间上的高维帕斯卡斜塔，而不是把二维剪切重新包装成独立模型。对任意 $q = 2^r$ 与 $d \geq 2$ ，维数跨越二的幂次时，公共周期按 $M = 2^{r+\lfloor \log_2 d \rfloor}$ 精确跳变。我们证明：只有当 M 整除迭代次数时，所有状态同时固定；其余时刻没有固定状态。由此得到完整的本原轨道数、动力学泽塔函数和有限库普曼行列式。截断多项式代换 $t \mapsto -t/(1 + t)$ 还给出保持原时钟的反演与反酉对称。证明依赖二项式系数的精确二进赋值，而有限枚举仅用于回归验证。本文不主张复杂性、目标谱匹配、算术局部结构或希尔伯特波利亚算子。关键词：仿射动力学；帕斯卡斜塔；二进赋值；精确周期；动力学泽塔；库普曼酉算子；时间反演。

1 Pascal ownership of every iterate

In $R = (\mathbb{Z}/q\mathbb{Z})[t]/(t^{d+1})$, encode a state by the unit

$$p_x(t) = 1 + x_d t + x_{d-1} t^2 + \dots + x_1 t^d. \quad (1)$$

Multiplication by $1 + t$ is exactly T , so

$$p_{T^n x} = (1 + t)^n p_x, \quad T^n x = x \iff q \mid \binom{n}{k} \quad (1 \leq k \leq d). \quad (2)$$

The last equivalence is independent of x : the unit p_x cancels in R . It already shows that every fixed set is either the whole state space or empty.

2 The sharp dyadic clock

Let $a = \lfloor \log_2 d \rfloor$ and $M = 2^{r+a}$. Since

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}, \quad v_2\left(\binom{n}{k}\right) \geq v_2(n) - v_2(k), \quad (3)$$

for $1 \leq k \leq n$. Coefficients with $k > n$ vanish and are automatically divisible by q . Thus $M \mid n$ implies divisibility by q for every $k \leq d$ without assigning a valuation to zero.

Conversely, $q \nmid n$ is detected by $k = 1$. If $q \mid n$ but $M \nmid n$, write $v_2(n) = r + b$ with $0 \leq b < a$ and choose

$$k = 2^{b+1} \leq 2^a \leq d. \quad (4)$$

The low $r + b$ binary digits of $n - 1$ are all one, and the low $b + 1$ digits of $k - 1$ are all one. Lucas reduction modulo two makes $\binom{n-1}{k-1}$ odd; hence

$$v_2\left(\binom{n}{k}\right) = r + b - (b + 1) = r - 1. \quad (5)$$

This coefficient violates (2). Therefore, for every parameter and iterate,

$$\text{Fix}(T^n) = \begin{cases} (\mathbb{Z}/q\mathbb{Z})^d, & M \mid n, \\ \emptyset, & M \nmid n. \end{cases} \quad (6)$$

This is the all-parameter proof; the 25,200 coefficient checks and 27,788 direct state periods packaged with C166 are finite regression sentinels.

3 Cycles, zeta, and the finite Koopman owner

Equation (6) gives least period M for all q^d states and exactly q^d/M primitive cycles. Thus

$$\log \zeta_T(z) = \sum_{n \geq 1} \frac{\# \text{Fix}(T^n)}{n} z^n = -\frac{q^d}{M} \log(1 - z^M), \quad \zeta_T(z) = (1 - z^M)^{-q^d/M}. \quad (7)$$

On $\mathcal{H} = \ell^2((\mathbb{Z}/q\mathbb{Z})^d)$ let $U_T f = f \circ T$. Its cycle decomposition immediately gives the same-clock identity

$$\det(I - zU_T) = (1 - z^M)^{q^d/M} = \zeta_T(z)^{-1}. \quad (8)$$

4 A truncated-ring reversor and scope

Nilpotence of t makes

$$\sigma(t) = -\frac{t}{1+t} = -t + t^2 - \dots + (-1)^d t^d \quad (9)$$

a well-defined substitution automorphism of R . Direct composition gives $\sigma^2(t) = t$ and $1 + \sigma(t) = (1 + t)^{-1}$; consequently

$$\sigma T \sigma = T^{-1}. \quad (10)$$

If $P_\sigma f = f \circ \sigma$ and J is complex conjugation, then $\Theta = P_\sigma J$ satisfies

$$\Theta^2 = I, \quad \Theta U_T \Theta^{-1} = U_T^{-1}. \quad (11)$$

The standalone shear is absorbed as $d = 2$; no product-rotation branch is presented as new progress. The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION). This finite source theorem supplies no complexity result, target trace/divisor/counting law, arithmetic local or Euler datum, root number, automorphy statement, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data availability. All claim-bearing evidence and programs are in the release; no external dataset is used. **Ethics.** No human participants, animals, or sensitive personal data are involved. **Author contributions/CRedit.** Anonymous technical certificate; Conceptualization, Formal analysis, Software, Validation, and Writing are recorded at package level. **Competing interests.** None known. **Funding.** No external funding is reported. **AI-use disclosure.** An AI coding assistant supported derivation planning, drafting, and code review. Packaged exact checks validate quantitative claims; the assistant was not treated as an external peer reviewer.